

Question Option C				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)		(i)	A Occipital lobe + B Frontal lobe (1) A Vision (1) B reasoning/planning/speech/movement/emotions/problem solving (1)	3			3		
			(ii)	EEG – measures {electrical/functional} activity of the brain (1) CT – gives brain images(1)	2			2		
	(b)		(i)	(During the critical period/between 0-5) synapses are formed and strengthened (1) If {Speech/Language} areas of the brain are not stimulated (1) There is more pruning of unused synapses (1) After critical period – brain is ‘hard wired’ and more difficult/impossible to form new synapses for language (1)		4		4		
			(ii)	{Less grey matter activity /darker scan } and fewer synapses			1	1		1
			(iii)	Any 1 from: high Cortisol levels (1) Epigenetic changes to the brain in the critical period/increased methylation(1) Maternal influence during pregnancy (1) e.g. stress/alcohol/smoking		1		1		

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	(c)		(i)	Hippocampus/temporal lobe	1			1		
			(ii)	Group 1 is rewarded every time – operant conditioning, there is a steady decrease in errors (1) Group 2, is latent learning until <u>day 10</u> (1) and then operant conditioning because reward given(1) Group 3 latent learning only no reward given (1)			4	4		
			(iii)	(-)70 % = 2 marks $\frac{20 - 6}{20} \times 100 = 1$ mark		2		2	2	
			(iv)	Any two (x1) from: Age/gender of rat (1) length of time left in the maze (1) mass of rat (1) same maze (1) same reward (1)		2		2		2
				Question 8 Option C total	6	9	5	20	2	3